

LANNION COTE DE GRANIT, France

WMO: 071180

Lat: 48.7553N

Lon: 3.4686W

Elev: 88

StdP: 100.27

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.3	0.0	-4.5	2.6	0.9	-3.1	3.0	1.8	21.2	11.0	18.5	10.3	2.8	150	0.593

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.3	24.7	18.5	22.7	17.7	21.2	17.1	19.4	23.1	18.5	21.5	17.7	20.2	4.0	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.2	20.6	17.3	12.5	19.7	16.7	12.0	19.1	55.6	23.2	52.6	21.4	50.3	20.4	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	10.6	9.3	DB	-3.8	29.1	1.6	3.0	-5.0	31.2	-6.0	32.9	-6.9	34.6	-8.1	36.7	(4)
(5)			WB	-4.3	21.7	1.8	1.5	-5.6	22.8	-6.6	23.7	-7.7	24.5	-9.0	25.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	6.8	6.6	8.0	9.7	12.1	14.8	16.6	16.4	15.2	12.8	9.5	7.4
	DBStd	4.45	2.95	2.82	2.70	2.55	2.43	2.36	2.07	2.09	2.51	2.72	2.57	3.04
	HDD10.0	453	102	98	73	36	7	1	0	0	1	7	38	90
	HDD18.3	2567	356	327	320	260	193	109	62	66	99	172	264	339
	CDD10.0	949	4	4	11	25	73	144	204	199	157	94	24	9
	CDD18.3	23	0	0	0	0	1	3	8	7	4	1	0	0
	CDH23.3	139	0	0	0	2	7	26	44	36	20	4	0	0
	CDH26.7	29	0	0	0	0	1	6	11	7	3	1	0	0
Wind	WSAvg	4.4	5.2	5.0	4.8	4.2	4.2	3.9	4.0	3.8	3.7	4.1	4.7	5.0
Precipitation	PrecAvg	862	93	84	57	67	58	51	51	55	46	88	104	109
	PrecMax	1117	170	162	162	173	115	118	118	147	135	191	204	236
	PrecMin	615	21	16	8	20	15	8	21	17	5	14	28	23
	PrecStd	115	43	40	33	37	26	28	25	29	30	40	46	48
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.6	15.7	18.4	21.6	23.9	26.8	27.9	27.1	26.1	22.1	17.0	14.8
		MCWB	12.1	11.3	12.6	15.0	17.9	19.4	20.0	19.5	18.8	17.8	15.0	12.6
	2%	DB	12.8	13.0	15.2	18.2	20.0	22.9	23.9	23.2	22.8	19.3	15.6	13.8
		MCWB	11.7	10.8	11.5	13.1	15.3	17.7	18.4	18.4	17.8	16.2	13.8	12.4
	5%	DB	12.1	11.9	13.3	15.7	18.0	20.7	22.1	21.3	21.0	17.9	14.2	12.9
		MCWB	10.9	10.3	10.9	12.0	14.3	16.6	17.5	17.6	17.1	15.5	12.8	11.8
	10%	DB	11.1	10.9	12.1	13.9	16.3	19.1	20.7	20.0	19.4	16.8	13.4	11.9
		MCWB	10.0	9.6	10.3	11.0	13.4	15.8	16.9	16.8	16.3	14.8	12.0	10.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.7	12.4	13.3	15.3	18.1	20.5	20.5	21.0	19.7	18.0	15.6	13.5
		MCDB	13.2	14.2	16.4	19.7	22.9	25.7	25.8	26.2	23.2	20.4	16.5	14.1
	2%	WB	12.0	11.3	12.2	13.6	15.7	18.3	19.1	19.3	18.6	16.9	14.1	12.7
		MCDB	12.7	12.6	14.2	17.4	18.8	21.5	22.8	22.3	21.5	18.8	15.1	13.4
	5%	WB	11.2	10.6	11.4	12.4	14.7	17.1	18.2	18.3	17.6	16.0	13.1	11.9
		MCDB	11.9	11.7	13.1	15.1	17.4	20.0	21.2	20.5	20.2	17.5	14.1	12.8
	10%	WB	10.3	9.7	10.5	11.5	13.8	16.1	17.4	17.5	16.8	15.2	12.2	11.1
		MCDB	11.0	10.8	12.0	13.6	16.0	18.4	19.9	19.4	18.8	16.5	13.2	12.0
Mean Daily Temperature Range	5% DB	MDBR	5.1	5.5	6.4	7.5	7.3	7.4	7.3	7.4	7.8	6.7	5.6	5.1
		MCDBR	5.3	7.1	8.7	10.6	10.1	10.3	10.1	9.7	10.1	7.9	6.5	5.3
		MCWBR	5.1	5.3	5.8	6.4	6.1	6.1	5.5	5.3	5.6	5.1	5.2	4.8
	5% WB	MCDBR	5.0	6.1	7.7	9.4	9.1	9.5	8.9	8.4	8.8	7.2	6.1	5.1
		MCWBR	5.1	5.2	5.5	6.1	5.8	6.0	5.4	5.2	5.4	5.0	5.3	5.0
Clear-Sky Solar Irradiance	taub	0.330	0.346	0.374	0.384	0.388	0.382	0.381	0.371	0.360	0.354	0.333	0.324	
	taud	2.413	2.393	2.326	2.316	2.352	2.395	2.400	2.449	2.460	2.467	2.444	2.378	
	Ebn at Noon	725	801	834	864	874	879	874	869	843	783	716	674	
	Edh at Noon	67	85	106	118	118	114	112	102	92	78	64	61	
All-Sky Solar Radiation	RadAvg	0.94	1.72	2.82	4.40	5.24	5.75	5.50	4.65	3.65	2.10	1.15	0.79	
	RadStd	0.08	0.20	0.29	0.41	0.42	0.62	0.33	0.37	0.27	0.17	0.11	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page